

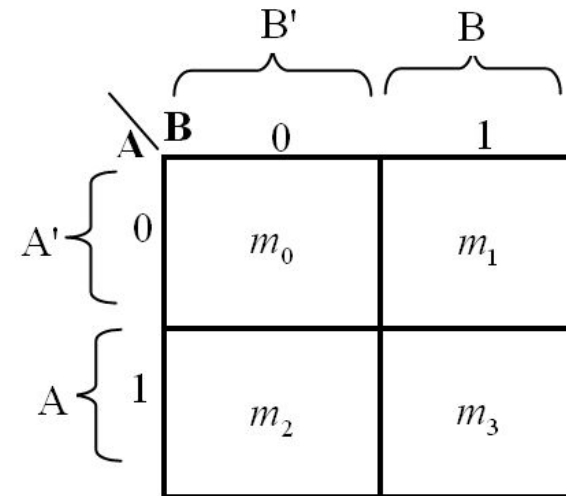
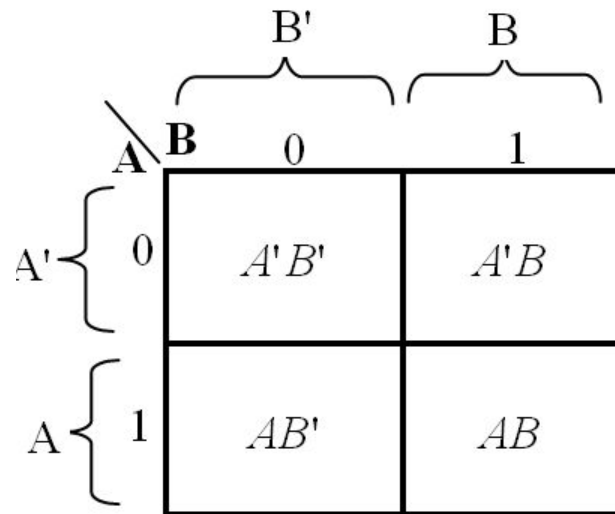
DIGITAL LOGIC DESIGN

- **K-map**

What is minimization?

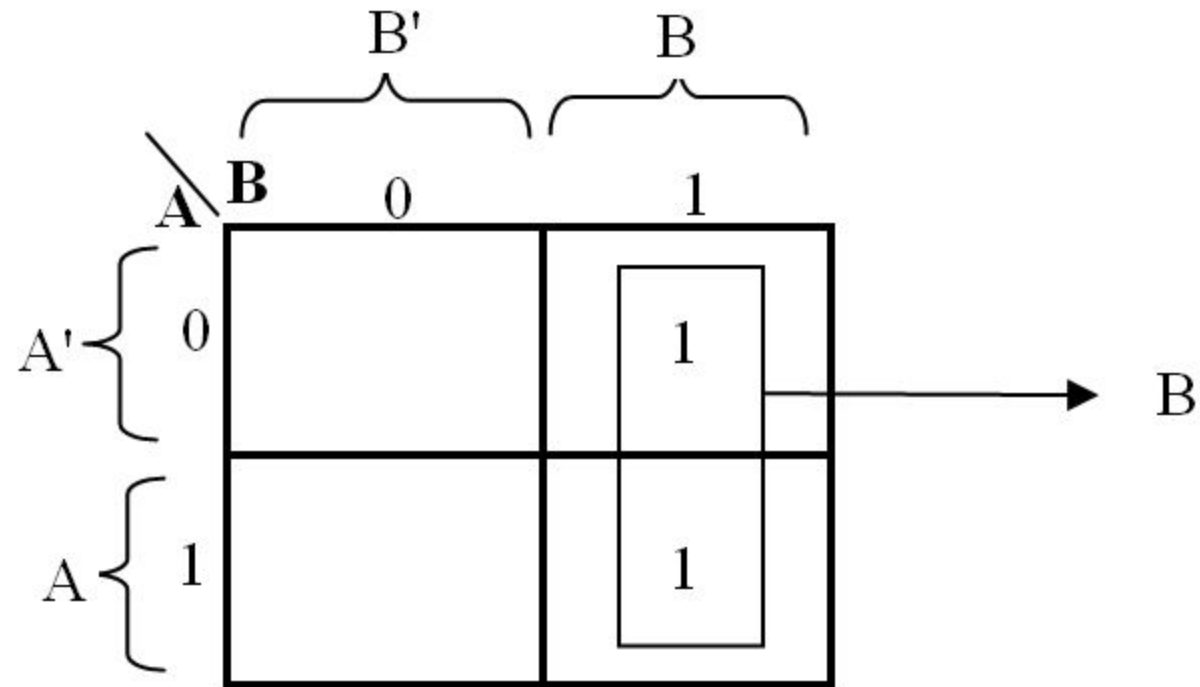
- Simplifying boolean expressions
- Algebraic manipulations is hard since there is not a uniform way of doing it
- Karnaugh map or K-map techniques is very commonly used

Two-Variable K-Map



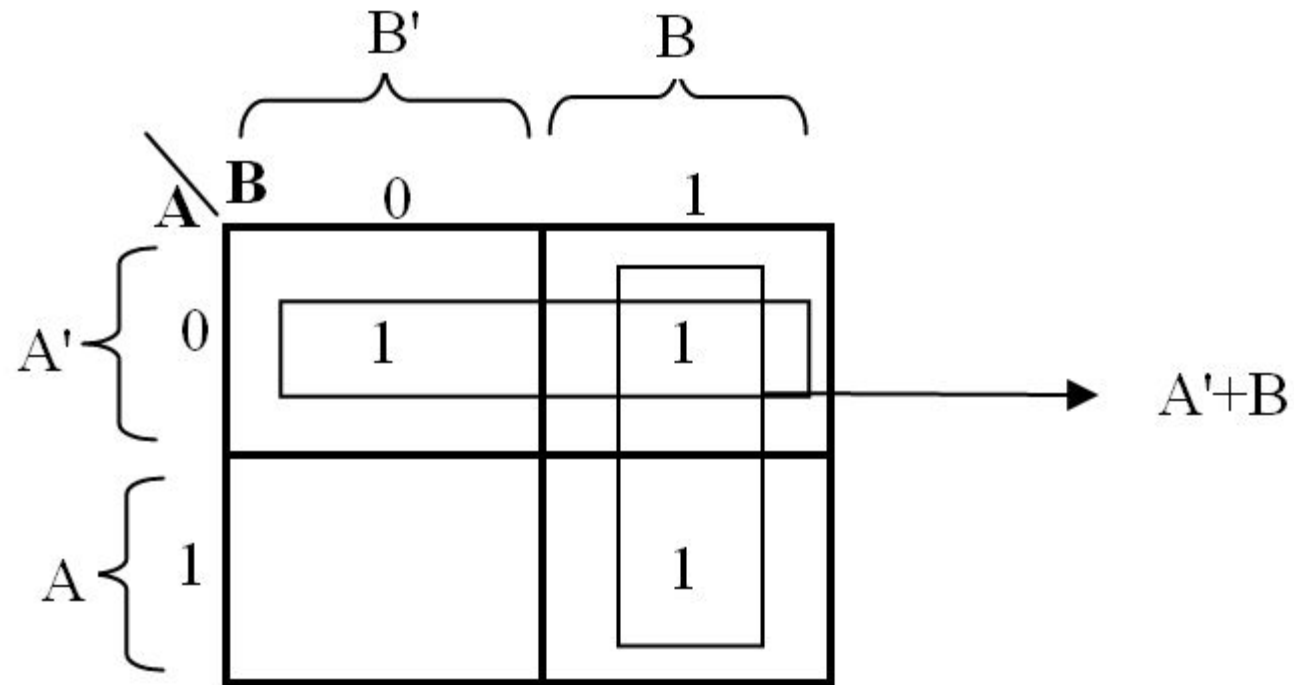
Example

$$F = AB + A'B = B$$



Example

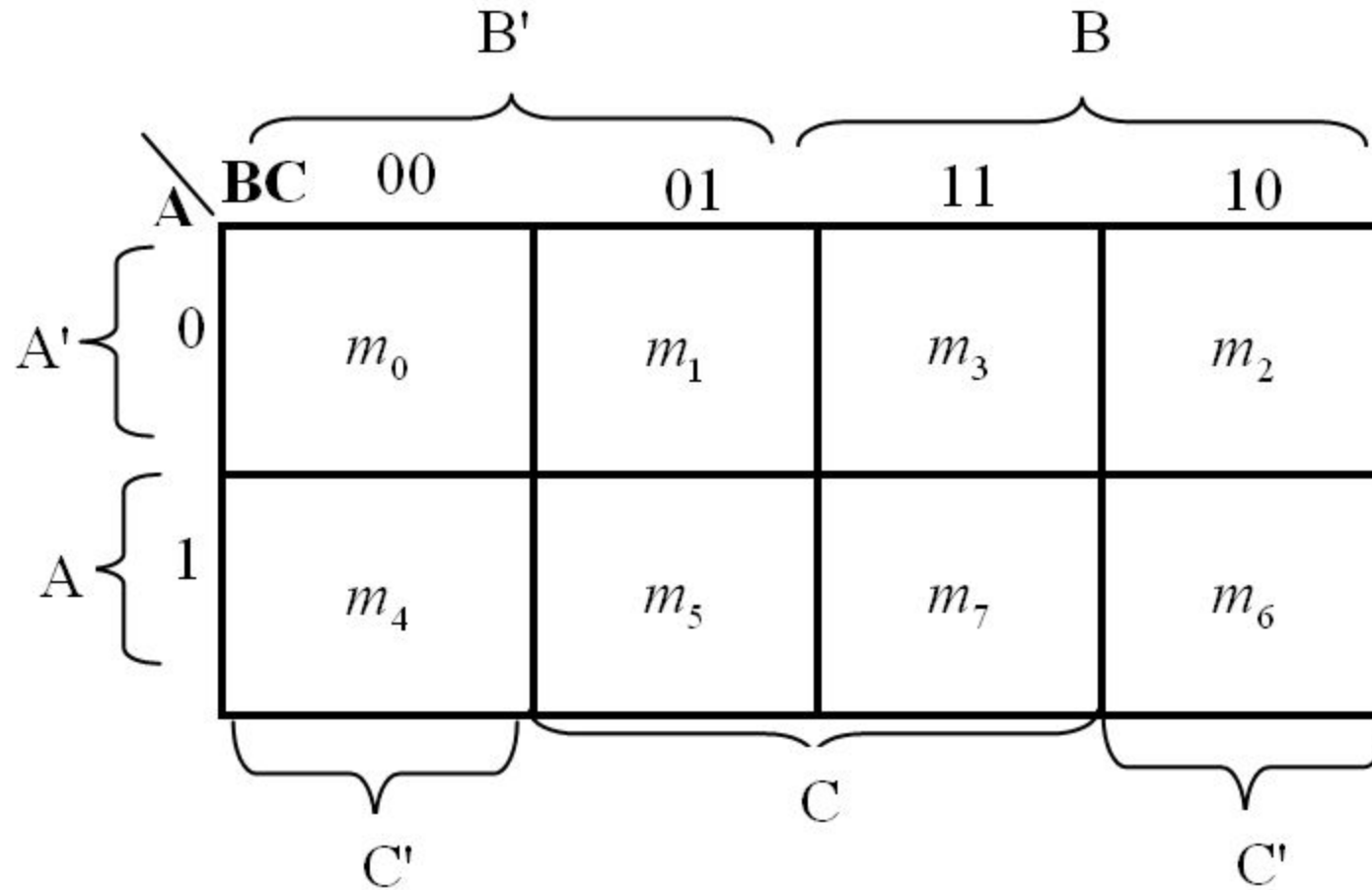
$$F = AB + A'B + A'B' = A' + B$$



Three-Variable K-Map

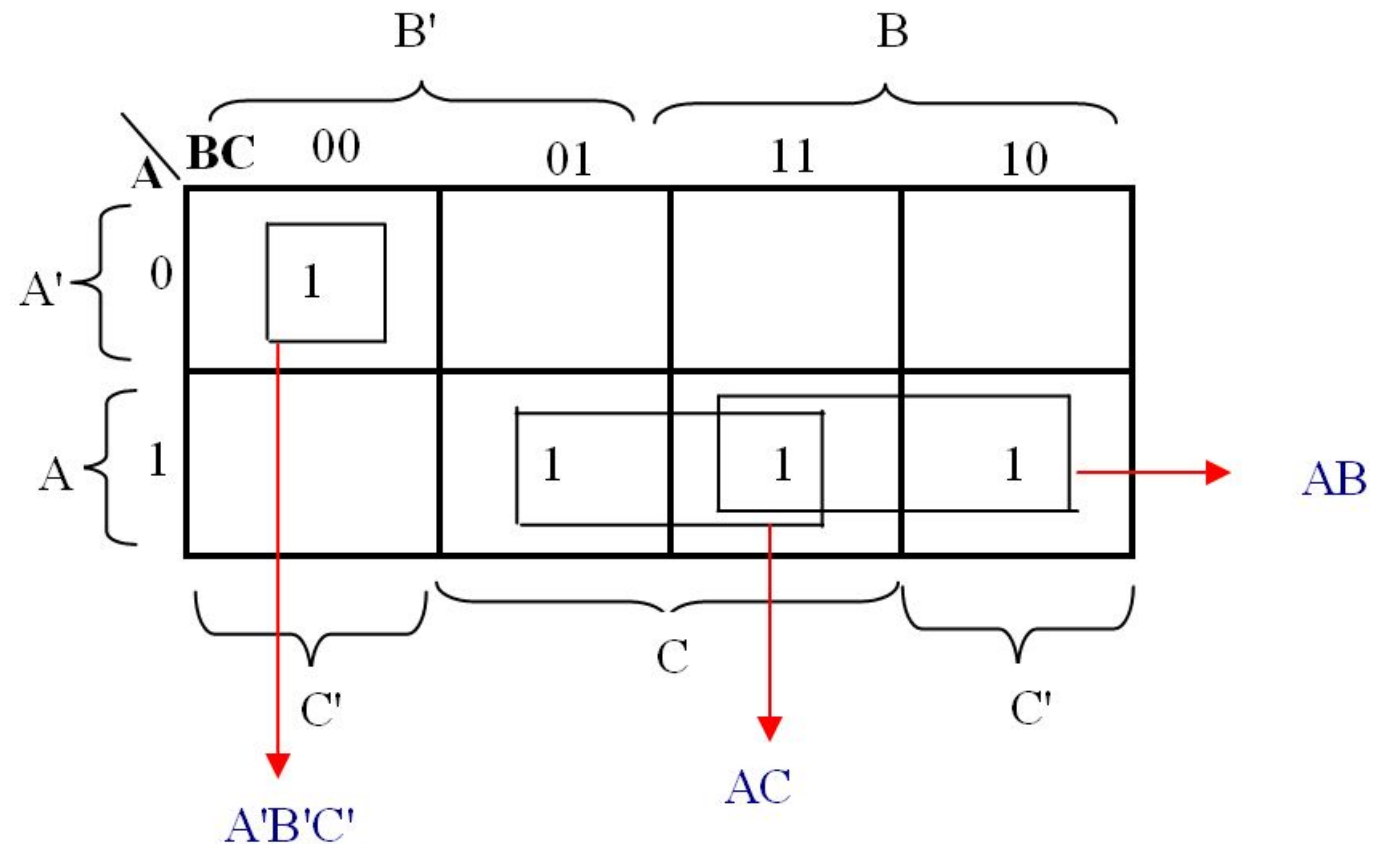
		B'		B	
		00		11	
		01		10	
A \ BC	BC	00	01	11	10
A' { 0		$A'B'C'$	$A'B'C$	$A'BC$	$A'BC'$
A { 1		$AB'C'$	$AB'C$	ABC	ABC'

Three-Variable K-Map



Example

$$F = ABC' + A'B'C' + AB'C + ABC$$
$$= AB + AC + A'B'C'$$



Note

- In K-maps, you can have groups of 2, 4, 8, or 16
- You cannot have groups of other combinations such as a group of 6

Quiz

- Simply the following Functions by using K-map

1. $F = A'B'C' + ABC' + A'BC' + AB'C$

2. $F = \sum (0, 2, 3, 5, 7)$

3.

x	y	z	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Example

- Represent F in the minimal format

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Example

- Represent F in the minimal format

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Example

$$F = \sum (0, 2, 3, 4, 5, 7)$$

- ◆ **Represent F in the minimal format**

Four-Variable K-Map

AB \ CD		C'		C	
		00	01	11	10
A'	00	$A'B'C'D'$	$A'B'C'D$	$A'B'CD$	$A'B'CD'$
	01	$A'BC'D'$	$A'BC'D$	$A'BCD$	$A'BCD'$
A	11	$ABC'D'$	$ABC'D$	$ABCD$	$ABCD'$
	10	$AB'C'D'$	$AB'C'D$	$AB'CD$	$AB'CD'$

The diagram illustrates the four-variable K-Map with groupings for variables A, B, C, and D.

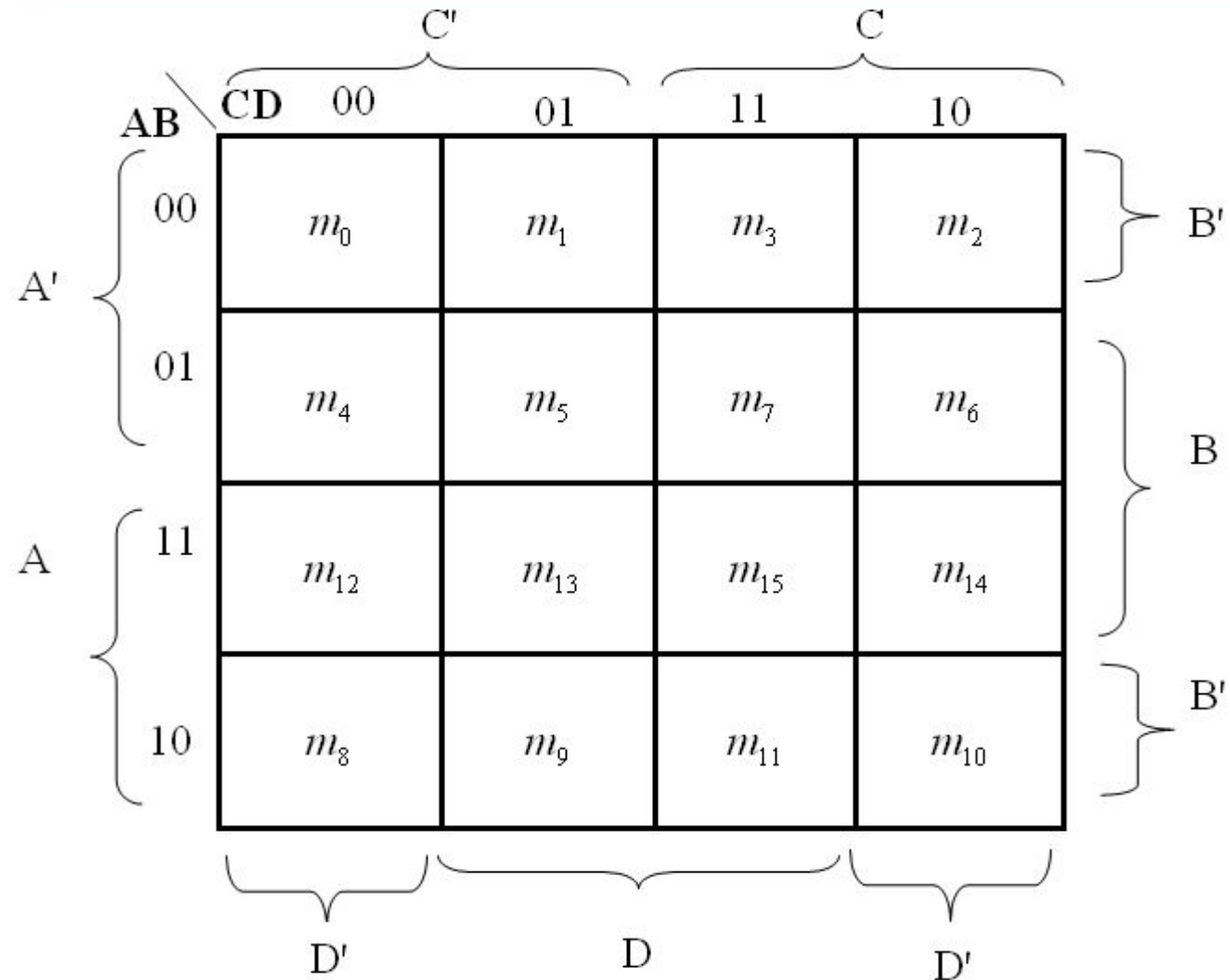
 - Variable A is grouped by rows: A' (rows 00, 01) and A (rows 11, 10).

 - Variable B is grouped by columns: B' (columns 00, 01) and B (columns 11, 10).

 - Variable C is grouped by pairs of columns: C' (columns 00, 01) and C (columns 11, 10).

 - Variable D is grouped by pairs of rows: D' (rows 00, 10) and D (rows 01, 11).

Four-Variable K-Map



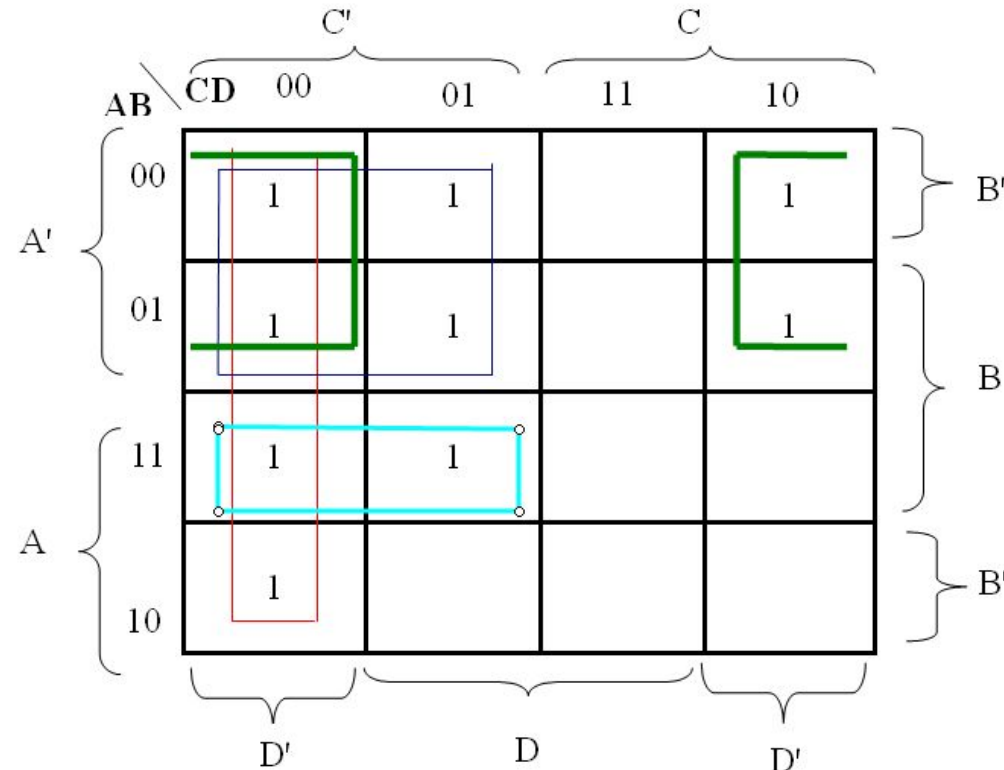
Gate-Level Minimization

Example

$$F(A, B, C, D) = \sum (0, 1, 2, 4, 5, 6, 8, 12, 13)$$

- ◆ Represent F in the minimal format and draw the network diagram

$$F(A, B, C, D) = A'C' + A'D' + C'D' + ABC$$



Example

$$F(A, B, C, D) = \sum (3, 5, 6, 8, 10, 11, 12, 13, 15)$$

- ◆ **Represent F in the minimal format and draw the network diagram**

Example

- Represent F in the minimal format and draw the network diagram

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Prime Implicants

- You must cover all of the minterms
- You must avoid redundancy
- You must follow some rules
- Prime Implicant
 - A product term that is generated by combining the maximum number of adjacent squares in the map
- Essential Prime Implicant
 - A minterm that is covered by only one prime implicant

Maxterm Simplification

- Remember

$$F = (F')'$$

Example

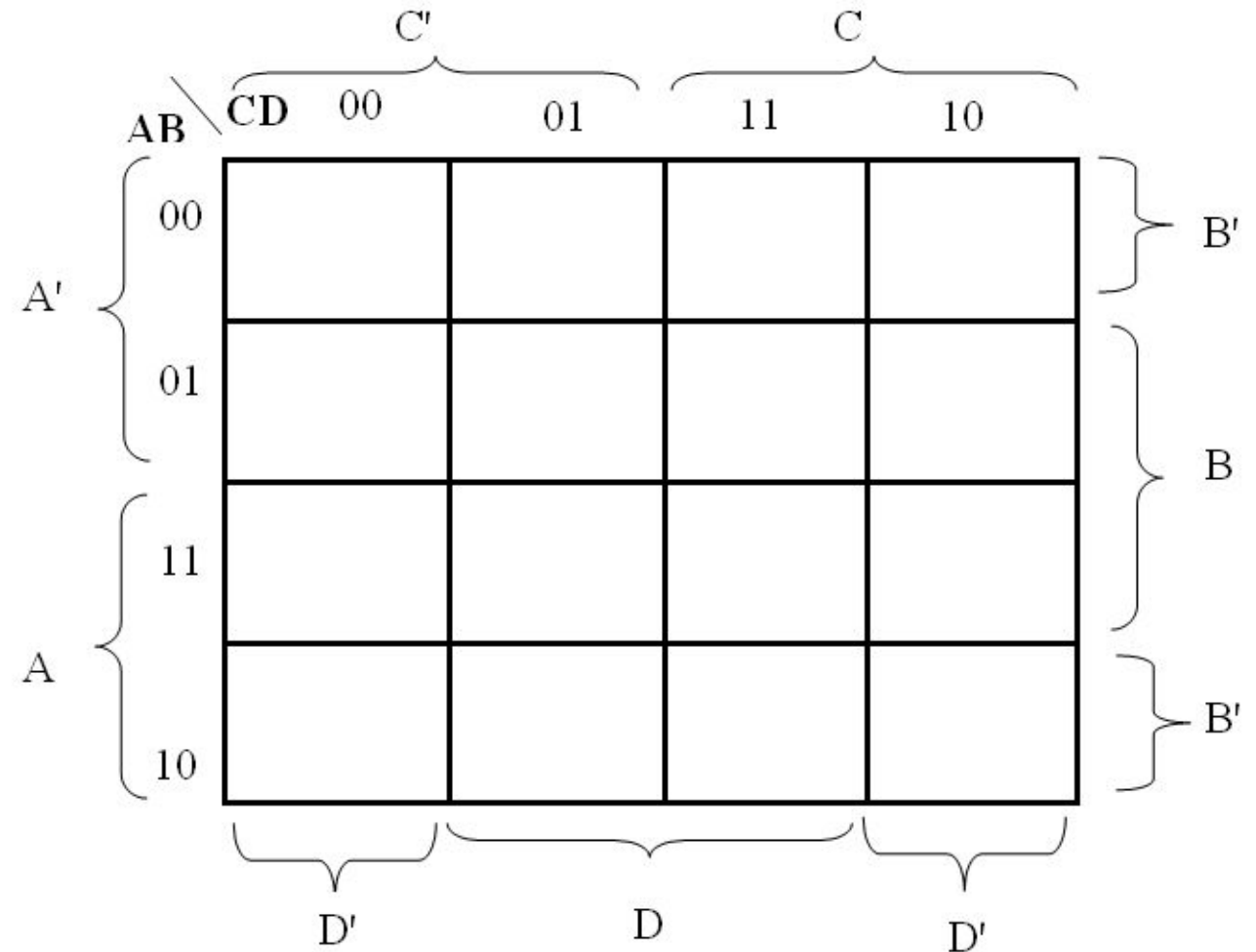
- Simplify F in product of sums (POS)

$$F(A, B, C, D) = \sum (1, 2, 3, 5, 6, 8, 10, 11, 12)$$

$$f(A, B, C, D) = \prod M(1, 5, 6, 12, 13, 14)$$

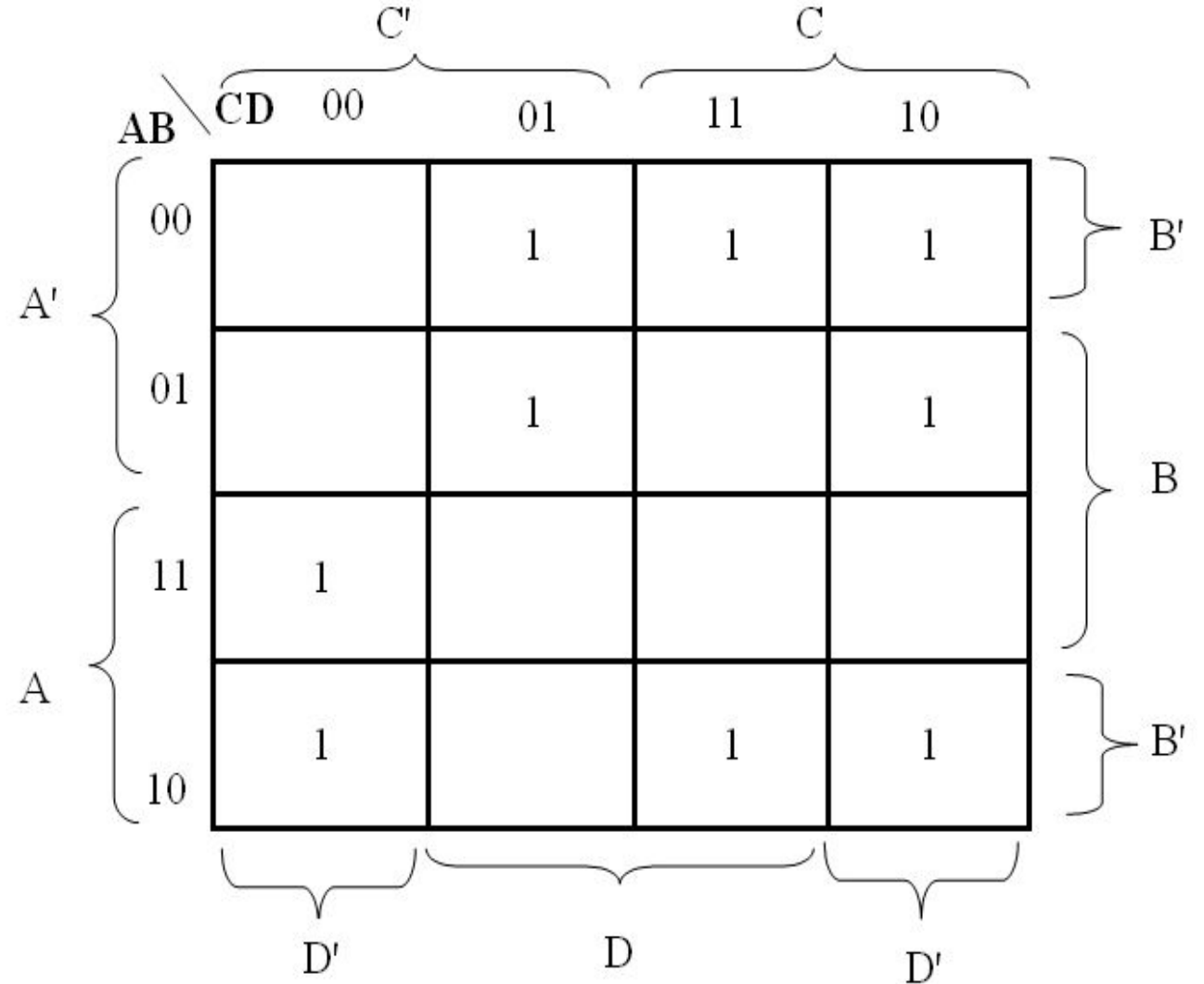
Example (cont)

- Step – 1
 - Fill the K-map for F



Example (cont)

- Step – 1
 - Fill the K-map for F



Example (cont)

- Step – 2
 - Fill zeros in the rest of the squares

AB \ CD		C'		C	
		00	01	11	10
A'	00	0	1	1	1
	01	0	1	0	1
A	11	1	0	0	0
	10	1	0	1	1
		D'		D	

Diagram illustrating a 4x4 Karnaugh map for a 4-variable function (A, B, C, D). The map is labeled with variables A, B, C, and D, and their complements (A', B', C', D'). The map shows the following values:

AB \ CD	00	01	11	10
00	0	1	1	1
01	0	1	0	1
11	1	0	0	0
10	1	0	1	1

The map is grouped into four 2x2 blocks, each labeled with a variable or its complement:

- Top-left block (A'B'): 00, 01, 11, 10
- Top-right block (A'B): 00, 01, 11, 10
- Bottom-left block (A'D'): 00, 01, 11, 10
- Bottom-right block (A'D): 00, 01, 11, 10

Example (cont)

- Step – 3
 - Cover zeros. This is your F'

AB \ CD		C'		C	
		00	01	11	10
A'	00	0	1	1	1
	01	0	1	0	1
A	11	1	0	0	0
	10	1	0	1	1
		D'		D	D'

Groupings on the right: {00, 01} is B', {01, 11} is B, {11, 10} is B'.

$$F(A, B, C, D)' = A' C' D' + A C' D + B C D + A B C$$

$\Rightarrow$

$$F(A, B, C, D) = (A + C + D)(A' + C + D')(B' + C' + D')(A' + B' + C')$$

Gate-Level Minimization

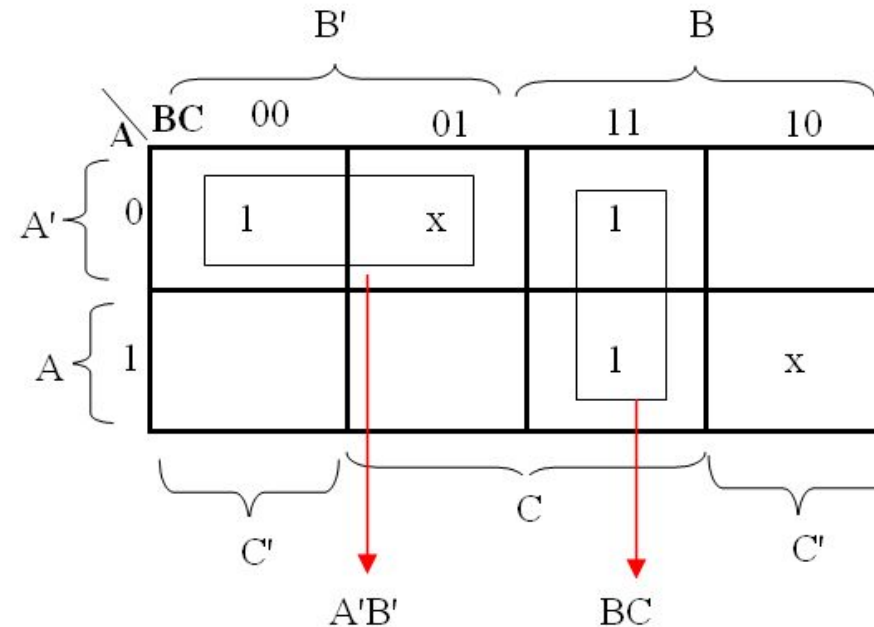
Don't Care Conditions

- The “Don't Care” conditions allow us to replace the empty cell of a [K-Map](#) to form a grouping of the variables which is larger than that of forming groups without don't care. While forming groups of cells, we can consider a “Don't Care” cell as 1 or 0 or we can also ignore that cell. Therefore, the “Don't Care” condition can help us to form a larger group of cells.

Don't Care Conditions

X can be considered as 0 or 1, whichever is more convenient

<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	1
0	0	1	x
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	x
1	1	1	1



$$F = A'B'C' + A'BC + ABC$$

$$F = A'B'C' + A'BC + A'B'C + ABC = A'B' + BC$$

$$F = A'B'C' + A'BC + A'B'C + ABC + ABC' = A'B' + BC + AB$$

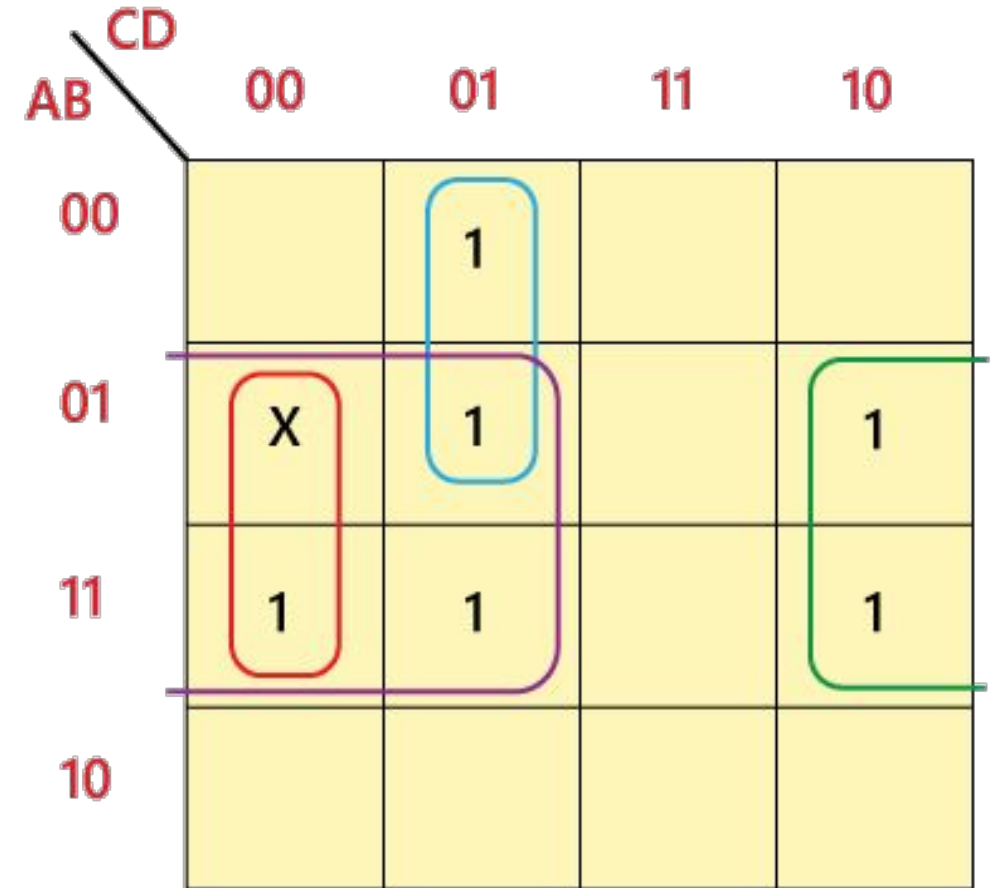
Example-1:

Minimize the following function in SOP minimal form using K-Maps:

$$F = m(1, 5, 6, 11, 12, 13, 14) + d(4)$$

Minimal Form:

$$F = BC' + BD' + A'C'D$$



Example-2:

Minimize the following function in SOP minimal form using K-Maps:

- $F(A, B, C, D) = m(1, 2, 6, 7, 8, 13, 14, 15) + d(3, 5, 12)$

- Minimal Form:

- $F = AC'D' + A'D + A'C + AB$

		CD			
AB		00	01	11	10
		00	1	X	1
	01		X	1	1
	11	X	1	1	1
	10	1			

Example 3:

Minimize $F(A,B,C,D) = m(0,1,2,3,4,5) + d(10,11,12,13,14,15)$ in POS minimal form

- The POS form of the given function is:
- $F(A,B,C,D) = M(6,7,8,9) + d(10,11,12,13,14,15)$

- Minimal Form:
- $F = A'(B' + C')$

AB \ CD	00	01	11	10
00				
01			0	0
11	X	X	X	X
10	0	0	X	X